Arunkarthick Ramesh

Mechanical Design Engineer — steel structures and lifting equipment

Lebenslauf · Karur, India



PERSÖNLICHE DATEN / Personal details

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PROFIL

I design load-bearing mechanical hardware and I close it with a calculation. For eight years that has meant a Creo model, a GD&T drawing, and a strength check by hand and in Ansys, against EN 81-20 and EN 1993. The parts are elevator shaft components, slings, and heavy-lift brackets: steel and sheet metal that must carry load, be weldable, and be releasable to production. That is the work I can do on heavy structures and mechanical installations.

WAS ICH EINBRINGE / What I can do for the team

- Turn a requirement and a standard into a concept, a detailed 3D model in Creo, and a manufacturing drawing with GD&T.
- Check steel members, welds, and bolts to EN 1993 myself: hand calculation plus Ansys static, modal, and buckling, including the EN 1993-1-8 combined tension and shear check.
- Treat brackets, slings, and lift frames as lifting structures. I have already analysed the Jeddah Tower elevator sling and a 12-ton heavy-lift bracket set.
- Write the strength note, material callout, and release pack, and stay with the job through PDM. I lead six strength engineers who work this way.
- Answer production from the drawing: sheet-metal tools, welding fixtures, and jigs, from three years in a chain factory.
- Use Siemens NX on the project if that is the office system. I learned NX and SolidWorks in my degree. Creo and AutoCAD are what I use in industry.

BERUFSERFAHRUNG / Professional experience

Special Engineering Specialist

12.2021 – heute · Chennai

KONE Elevators · Shaft Mechanical & Component Engineering

- Design EN 81-20 load-bearing shaft components in Creo for European and Chinese elevators, about three projects a month: model, interference check, drawing.
- Check the same parts and the large assemblies in Ansys Workbench and by manual EN 1993 calculation for the member, the weld, and the bolt.
- Two months on site in Finland on high-rise elevator design and structural analysis with the European engineering team.
- Lead a six-person strength team from the input data to PDM release. Use 5 Whys when a design fault comes back. Report status to senior stakeholders.
- Six Sigma Green Belt initiative on this work: engineering time from 50 hours to 3 hours per project, cost from about €1,000 to €120, and a further 30% cost reduction. With R&D, analysis time from 18 hours to 8 hours.
- Ansys Mechanical 2025 R1, IronPython: read every beam probe (tension from axial force, shear from the larger end, bolt size from the beam radius) and selected joint probes, then check slip and EN 1993-1-8 tension, shear, and combined utilization. The script writes the legend and the PDF into the strength file.
- Jeddah Tower, about 120 hours: structural analysis of the elevator sling, lifting configuration, and safety of that steel assembly.
- King Salman project, about 30 hours: structural analysis of a 12-ton heavy-duty lift, including the strength of the heavy brackets.

Design Engineer
04.2019 – 12.2021 · Dindigul

Renold Chain India · also Renold Einbeck, Germany

- Released 20–25 manufacturing drawings a month in AutoCAD and Creo, through Windchill, for new chain parts, legacy conversion, and revisions.
- Designed the shop hardware around those parts: sheet-metal forming tools, stamping dies, welding fixtures, and production jigs.
- Modelled segmented sprockets, hub sprockets, and the large drive assemblies they sit in.
- Prepared CSR, BOM, and EDS, and put heat treatment and metallurgy on the drawing set. Product standards: ISO 606, ASME B29.8, BS 4116.

Business Development Engineer
01.2019 – 04.2019 · Dindigul

Renold Chain India

- Matched OEM and dealer applications to the chain range, with regional sales. Customer reach increased 20%.

Graduate Engineer Trainee
07.2018 – 12.2018 · Dindigul

Renold Chain India

- Industrial training in the chain plant: design office and manufacturing of the product I later drew.

AUSBILDUNG / Education

B.E. Mechanical Engineering · CGPA 7.5 / 10
2014 – 2018 · Coimbatore

Sri Ramakrishna Institute of Technology, Anna University

- Academic CAD only, not used in my KONE or Renold work: Siemens NX and SolidWorks, for parts, assemblies, and drawings.
- Member, SAE. Member, The Institute of Indian Foundrymen.

KENNTNISSE / Skills

Industrie-CAD	Creo 9 (sheet metal, assemblies, Pro/Program), AutoCAD
Studium-CAD	Siemens NX, SolidWorks — degree only, not industrial projects
FEA	Ansys Workbench and Mechanical: static, modal, buckling, mesh convergence; HyperMesh
Normen	EN 81-20, EN 1993 including bolts and welds, ISO 606, ASME B29.8, BS 4116
Freigabe	GD&T, Windchill / PDM, MathCAD, strength reports, BOM, EDS

SPRACHEN / Languages

Englisch	Professional working language
Tamil	Native
Deutsch	A1

ZERTIFIKATE / Certificates

- Six Sigma Yellow Belt — KONE. The Green Belt initiative above was a project, not this certificate.
- HyperMesh for FEA, plastics and sheet metal — Skill-Lync, May 2023.
- Geometric Dimensioning & Tolerancing — Udemy, May 2020.
- Sheet Metal Design — Udemy, March 2020.
- Innovation Award 2026, top 10 of 169 ideas. Ace Performer, KONE, 2023 and 2025.